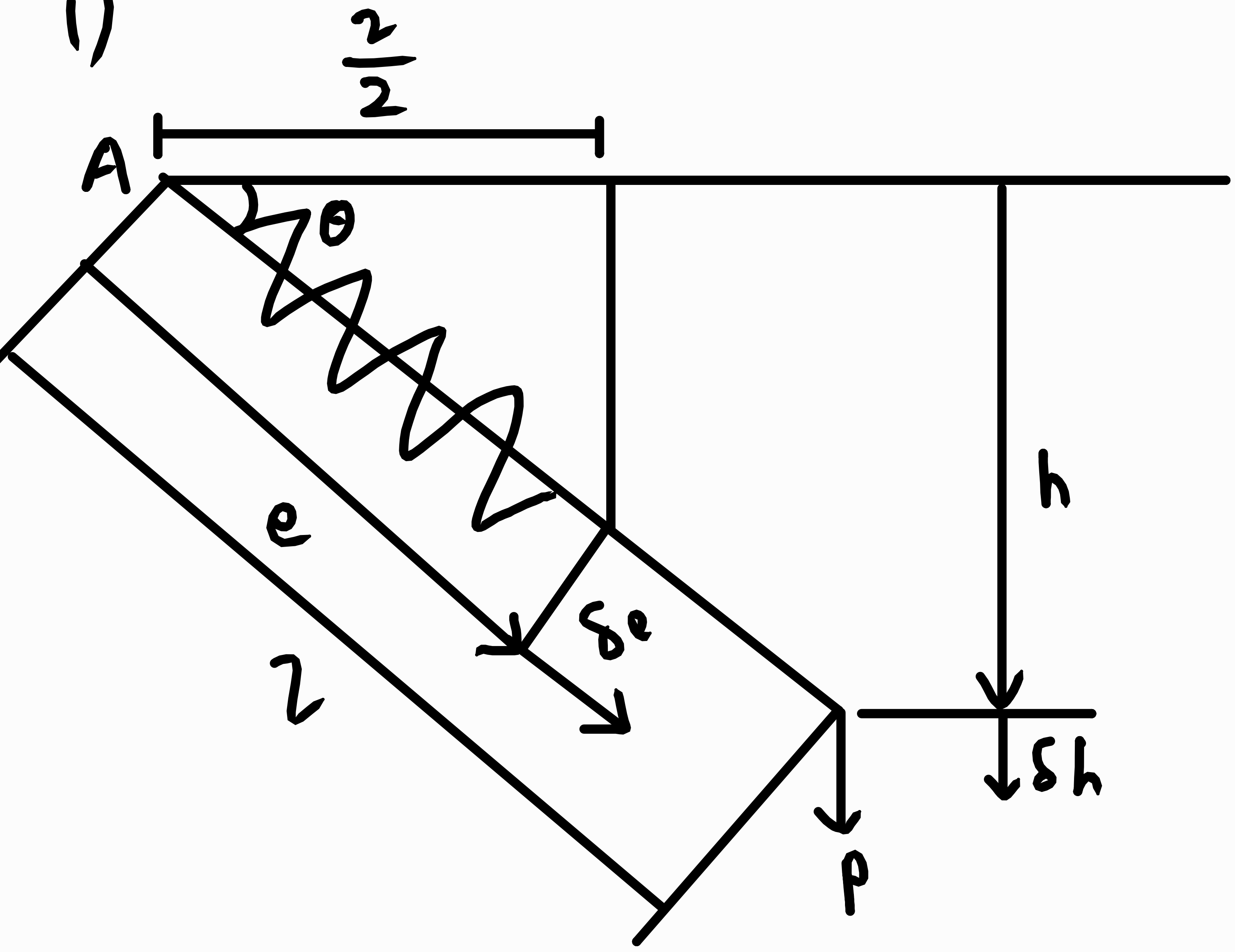
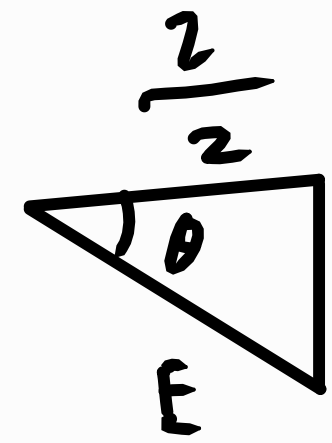
1)





$$E\cos\theta = \frac{l}{2}$$

$$E = \frac{l}{2\cos\theta}$$

$$h = l\sin\theta$$

$$e = \frac{l}{2\cos\theta} - \frac{l}{2}$$

$$\delta h = \frac{dh}{d\theta}\delta\theta$$

$$= l\cos\theta\,\delta\theta$$

$$\delta e = \frac{de}{d\theta}\delta\theta$$

$$= \frac{l}{2}\sec\theta\tan\theta\,\delta\theta$$

$$\delta W = \delta U$$

$$P\delta h = F\delta e$$

$$P(\cancel{l}\cos\theta\,\cancel{\delta\theta}) = F\left(\frac{\cancel{l}}{2}\sec\theta\tan\theta\,\cancel{\delta\theta}\right)$$

$$P = \frac{1}{2}F\sec^2\theta\tan\theta$$

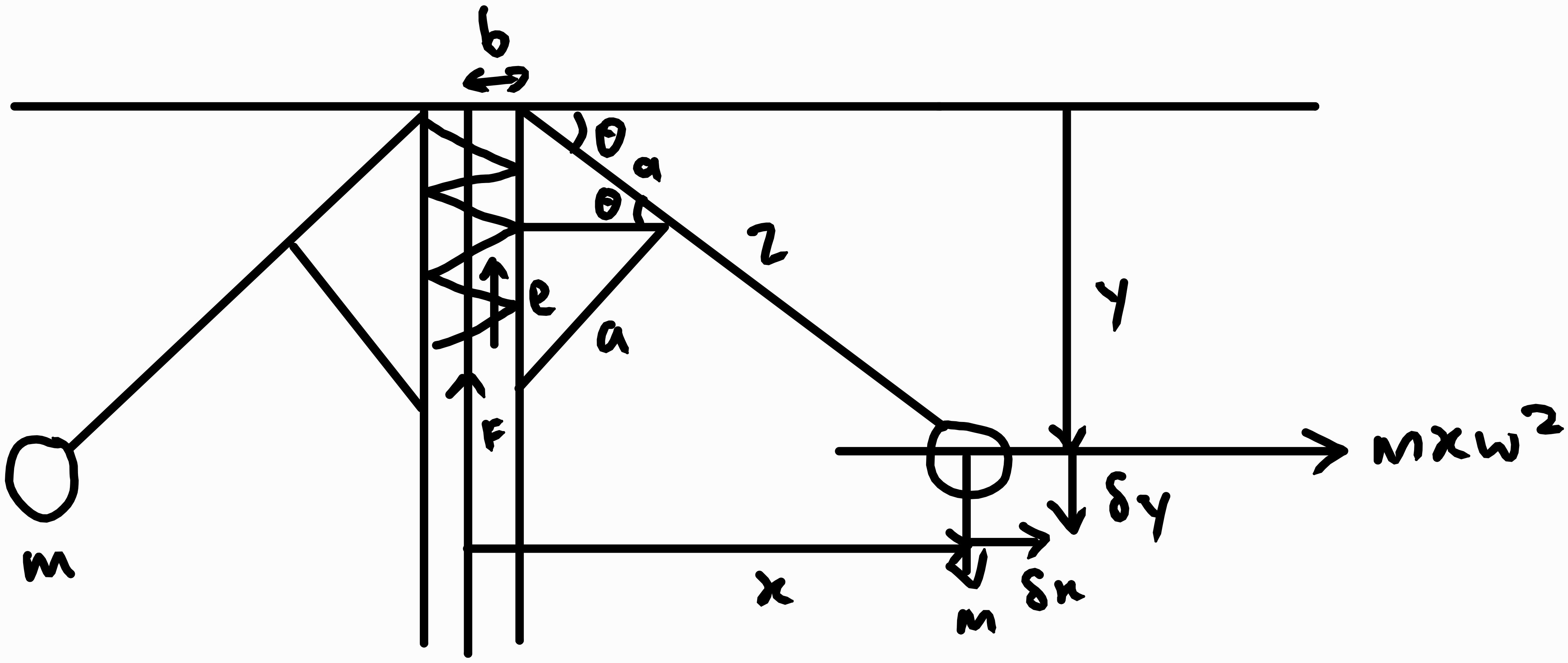
1) $P = \frac{1}{2} F \sec^2\theta \tan\theta$

$P = \frac{1}{2} ke \sec^2\theta \tan\theta$

$= \frac{1}{2} k \sec^2\theta \tan\theta \left(\frac{l}{2}\sec\theta - \frac{l}{2}\right)$

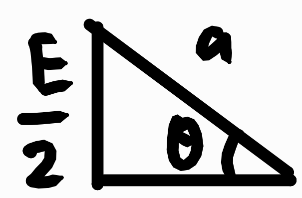
$= \frac{1}{4} kl \sec^3\theta \sin\theta (\sec\theta - 1)$

2)



$$x = b + l\cos\theta$$

$$e = 2a(1-\sin\theta)$$



$$\frac{E}{2} = a\sin\theta$$

$$E = 2a\sin\theta$$

$$y = l\sin\theta$$

$$\delta x = -l\sin\theta\,\delta\theta \qquad \delta e = -2a\cos\theta\,\delta\theta$$

$$\delta y = l\cos\theta\,\delta\theta$$

$$\delta W = \delta U$$

$$2mg\,\delta y + 2mxw^2\,\delta x = F\,\delta e$$

$$\cancel{2}mg(l\cos\theta\cancel{\delta\theta}) + \cancel{2}m(b+l\cos\theta)w^2(-l\sin\theta\cancel{\delta\theta}) = F(-\cancel{2}a\cos\theta\cancel{\delta\theta})$$

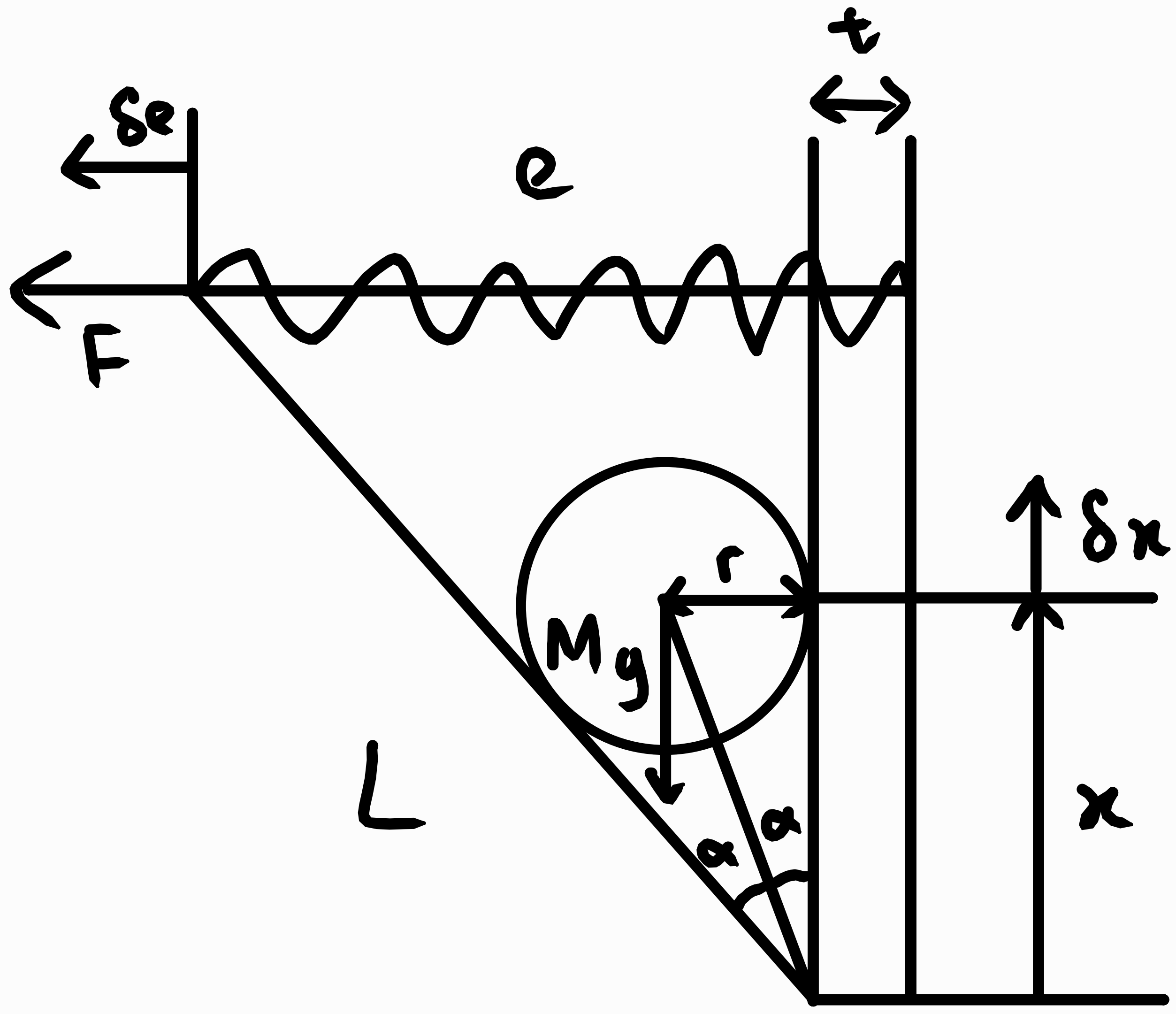
$$mgl\cos\theta - mw^2 l\sin\theta(b+l\cos\theta) = -kea\cos\theta$$

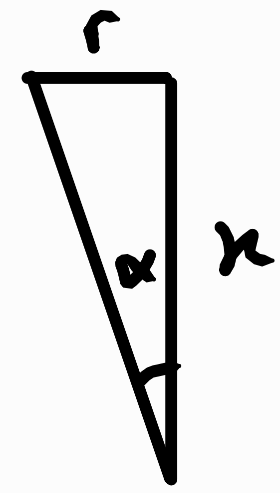
$$mw^2 l\sin\theta(b+l\cos\theta) - mgl\cos\theta = k(2a)(1-\sin\theta)a\cos\theta$$

$$w^2 ml\sin\theta(b+l\cos\theta) - mgl\cos\theta = 2a^2k(1-\sin\theta)\cos\theta$$

$$w = \sqrt{\frac{2a^2k(1-\sin\theta)\cos\theta + mgl\cos\theta}{ml\sin\theta(b+l\cos\theta)}}$$

3)





$$\tan\alpha = \frac{r}{x}$$

$$x = \frac{r}{\tan\alpha}$$

$$e = L\sin 2\alpha + t - l_0$$

$$x = \frac{r}{\tan\alpha} = r\cot\alpha$$

$$\delta e = 2L\cos 2\alpha\,\delta\alpha \qquad \delta x = -r\,\mathrm{cosec}^2\alpha\,\delta\alpha$$

$$\delta W = \delta U$$

$$-Mg\,\delta x = F\delta e$$

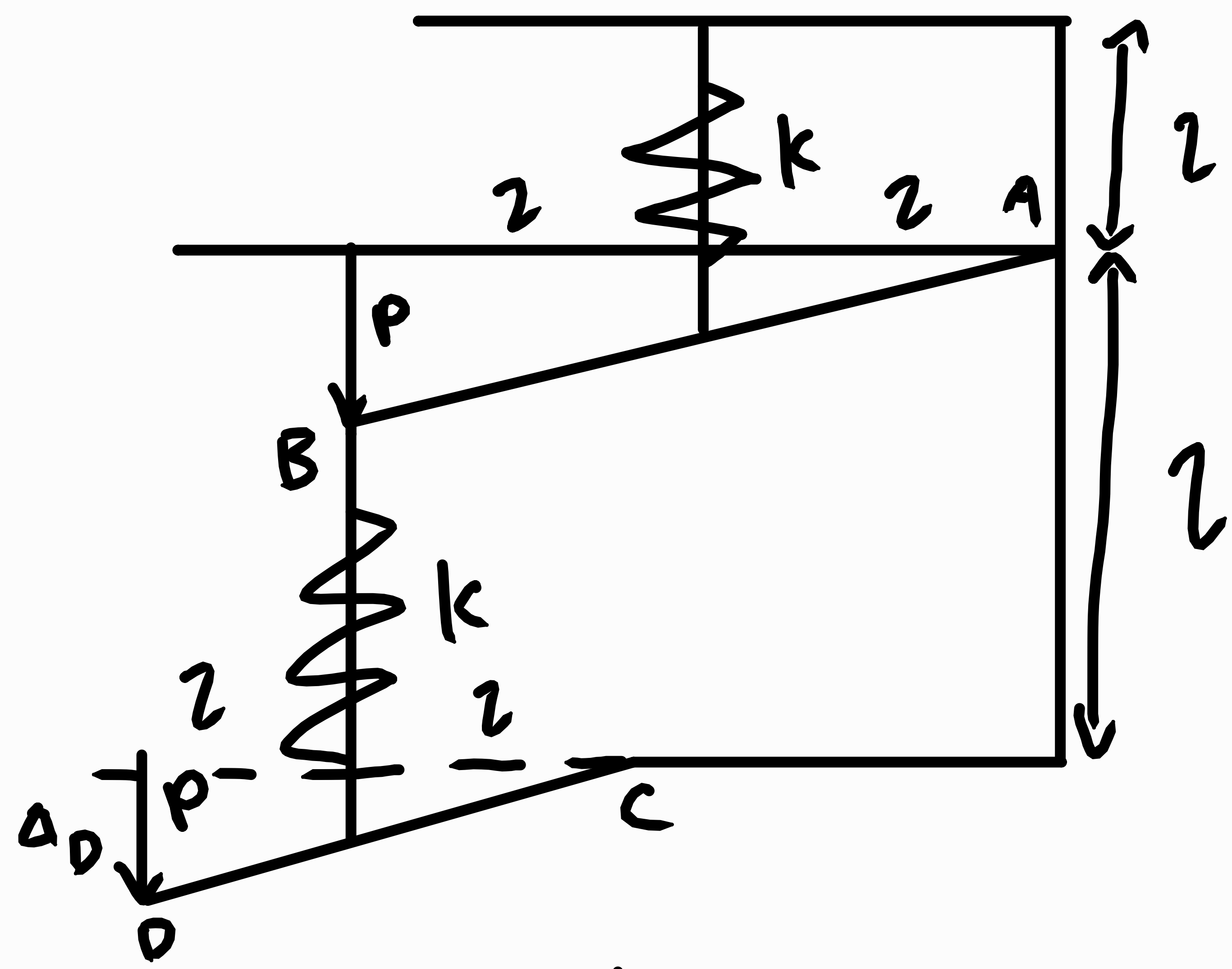
$$-Mg(-r\,\mathrm{cosec}^2\alpha\,\cancel{\delta\alpha}) = F(2L\cos 2\alpha\,\cancel{\delta\alpha})$$

$$\frac{Mgr}{\sin^2\alpha} = 2KeL\cos 2\alpha$$

$$\frac{Mgr}{\sin^2\alpha} = 2K(L\sin 2\alpha + t - l_0)L\cos 2\alpha$$

$$K = \frac{Mgr}{2L\cos 2\alpha\sin^2\alpha(L\sin 2\alpha + t - l_0)}$$

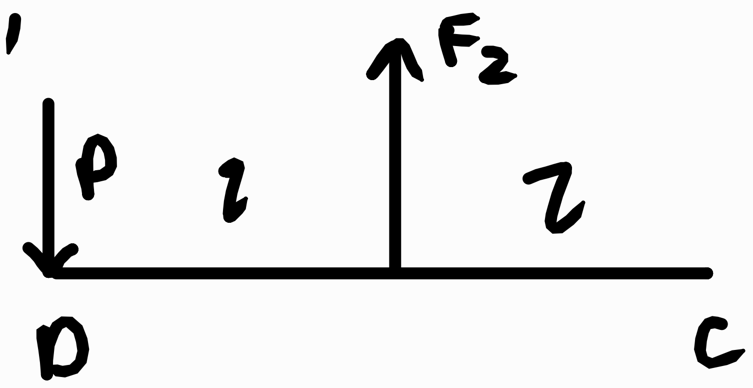
4)



$$\delta W^* = \delta U^*$$

$$\Delta_D \, \delta P_D = \frac{F_1}{k}\delta F_1 + \frac{F_2}{k}\delta F_2$$

On bar DC,

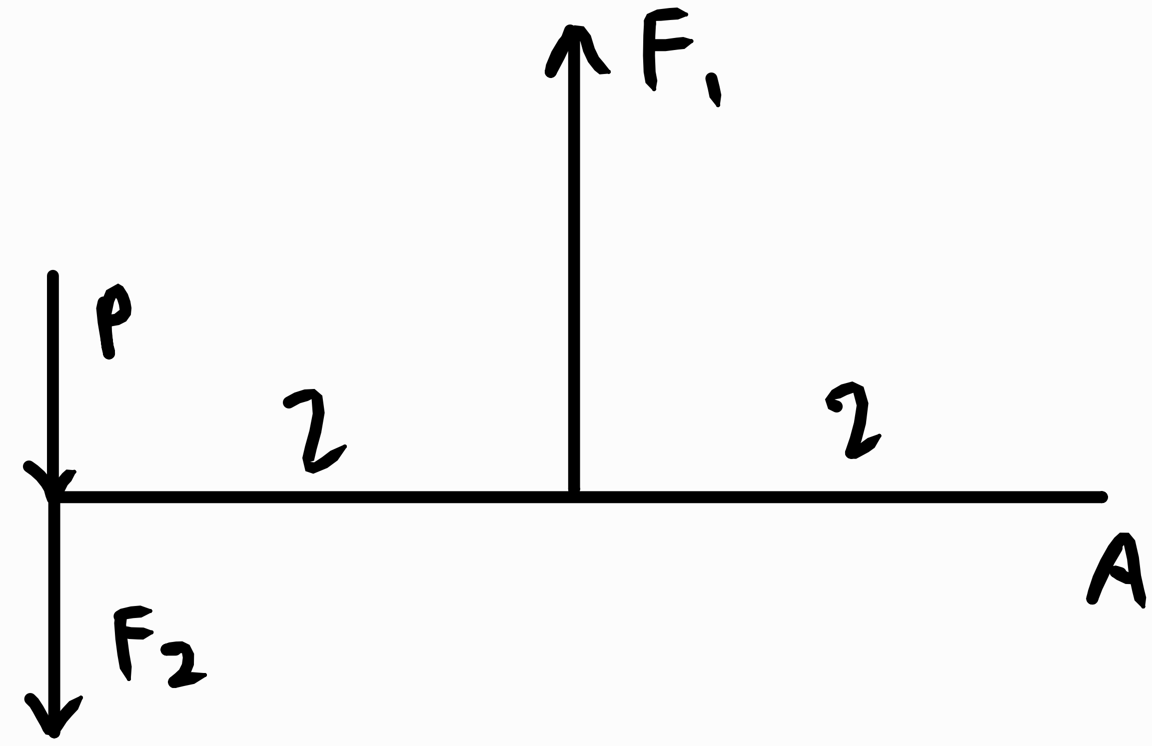


$$\Sigma M_c = 0$$

$$P(2\cancel{l}) = F_2 \cancel{l}$$

$$F_2 = 2P$$

4) On bar AB,

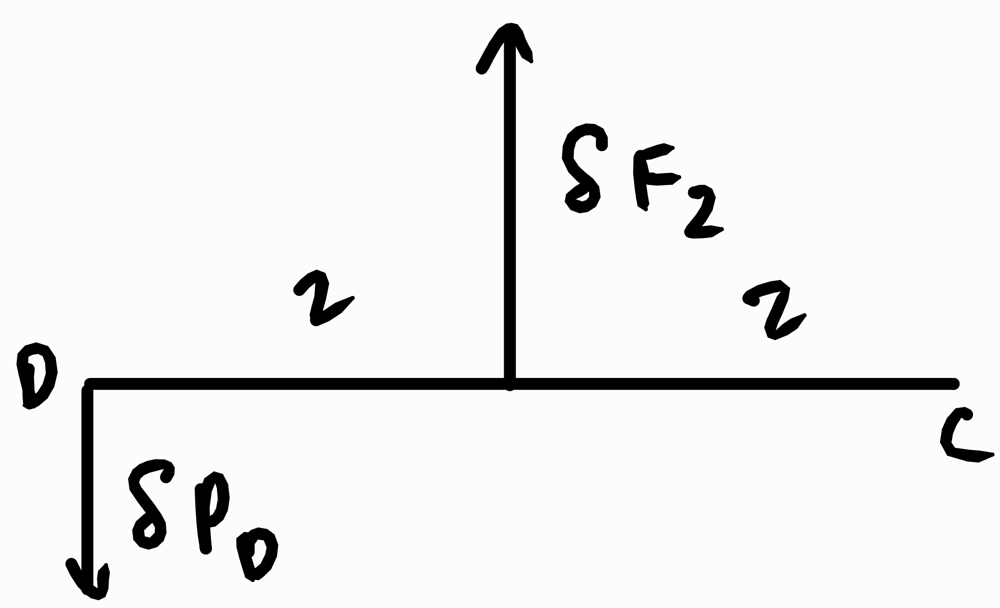


$$(P+F_2)(2\cancel{x}) = F_1\cancel{x}$$

$$(P+2P)(2) = F_1$$

$$F_1 = 6P$$

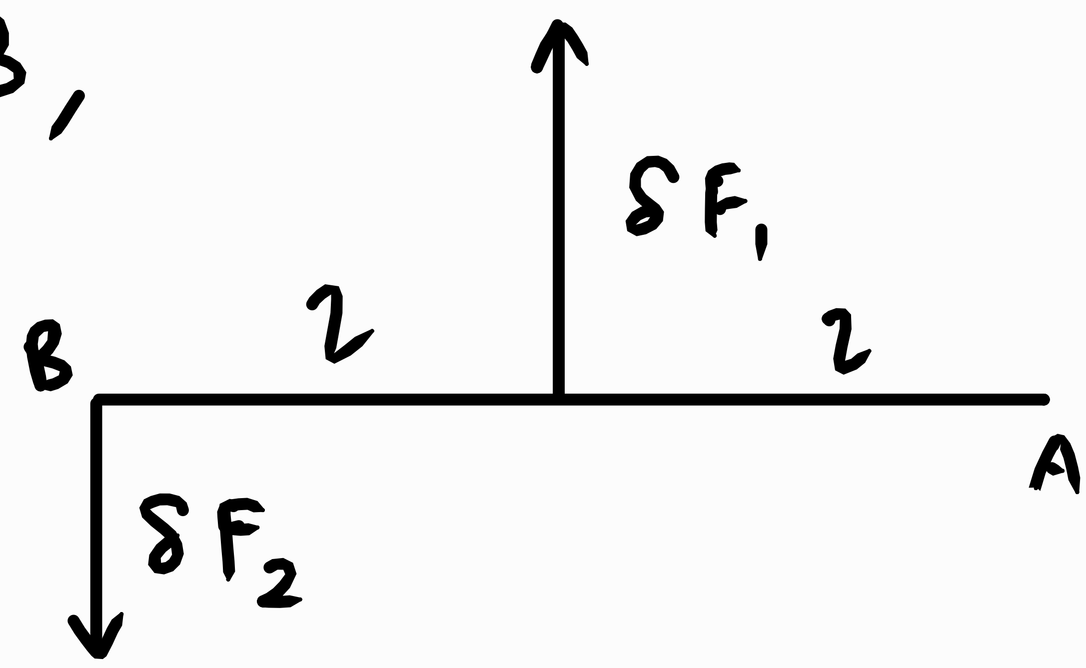
On bar CD,



$$\delta P_D(2\cancel{x}) = \delta F_2\cancel{x}$$

$$\delta F_2 = \delta P_D$$

4) On bar AB,



$$\delta F_2 (2\cancel{x}) = \delta F_1 (\cancel{x})$$

$$2\delta F_2 = \delta F_1$$

$$\delta F_1 = 4\delta P_D$$

$$\therefore F_1 = 6P, \quad F_2 = 2P, \quad \delta F_1 = 4\delta P_D, \quad \delta F_2 = 2\delta P_D$$

$$\Delta_D \delta P_D = \frac{F_1}{k}\delta F_1 + \frac{F_2}{k}\delta F_2$$

$$\Delta_D \cancel{\delta P_D} = \frac{6P}{k}(4\cancel{\delta P_D}) + \frac{2P}{k}(2\cancel{\delta P_D})$$

$$\Delta_D = \frac{28P}{k}$$